3 **(a)** Sketch the graph of $y = \arctan x$ where x is in radians. **[2]**

(b) **In this question you must show detailed reasoning.**

Find all points of intersection of the curves $y = 3\sin x \cos x$ and $y = \cos^2 x$ for $-\pi \leqslant x \leqslant \pi$. **[6]**

3(a)

3(b)